Sample Complexity Bounds

(i) $P(A_1 \cup \cdots \cup A_k) \le \sum_i P(A_i)$

(ii) $Z_1, \ldots, Z_n \overset{i.i.d}{\sim} \text{Bernoulli}(\phi)$

and $\hat{\phi} = \frac{1}{n} \sum Z_i$ then $P(|\phi - \hat{\phi}| > \gamma) \le \frac{2}{e^{2\gamma^2 n}}$

---

training error $\hat{\varepsilon}(h) = \frac{1}{n} \sum 1\{h(x^{(i)}) \neq y^{(i)}\}$

generalization error $\varepsilon(h) = P_{(x,y)\sim p}(h(x) \neq y)$

Note. Same data distribution!

Hypothesis Class $\mathcal{H} = \{h_1, \ldots, h_k\}$

- $P(|\varepsilon(h_i) - \hat{\varepsilon}(h_i)| > \gamma) \le 2e^{-2\gamma^2 n}$
- $P(\exists h \in \mathcal{H} \; |\varepsilon(h) - \hat{\varepsilon}(h)| > \gamma) \le 2ke^{-2\gamma^2 n}$
- $P(\forall h \in \mathcal{H} \; |\varepsilon(h) - \hat{\varepsilon}(h)| \le \gamma) \ge 1 - 2ke^{-2\gamma^2 n}$

---

with prob $\ge 1-\delta$

$$\varepsilon(\hat{h}) \le \varepsilon(h^*) + 2\sqrt{\frac{1}{2n} \log \frac{2k}{\delta}}$$

$\hat{h}$: training 최적화

h^* : generalization 최적화

$\because \; \varepsilon(\hat{h}) \le \hat{\varepsilon}(\hat{h}) + \gamma \le \hat{\varepsilon}(h^*) + \gamma \le \varepsilon(h^*) + 2\gamma$

---

for $\varepsilon(\hat{h}) \le \varepsilon(h^*) + 2\gamma$ to hold with prob $\ge 1-\delta$

$\because \; \sqrt{\frac{1}{2n} \log \frac{2k}{\delta}} \le \gamma$

$$n \ge \frac{1}{2\gamma^2} \log\left(\frac{2k}{\delta}\right) = O\left(\frac{1}{\gamma^2} \log \frac{k}{\delta}\right)$$

Bias-Variance Tradeoff

increase $|\mathcal{H}|$ then

(i) $\varepsilon(h^*)$ 감소

(ii) $\varepsilon(\hat{h}) \le \varepsilon(h^*) + 2\sqrt{\frac{1}{2n} \log \frac{2k}{\delta}}$ 증가

# of parameters

$$h_\theta(x) = \mathbb{1}\{\theta_0 + \theta_1 x + \cdots + \theta_d x_d \geq 0\}$$

$\{u_0^2 - v_0^2\}$ $\{u_1^2 - v_1^2\}$ $\{u_d^2 - v_d^2\}$

- $\mathcal{H}$ shatters $S = \{x^{(1)} \cdots x^{(D)}\}$: any labels $\{y^{(1)} \cdots y^{(D)}\}$

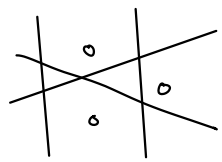
  $\exists\, h \in \mathcal{H}$ s.t $h(x^{(i)}) = y^{(i)}$

- Vapnik – Chervonenkis Dimension

  $VC(\mathcal{H})$: size of largest set that is shattered by $\mathcal{H}$

e.g $\mathcal{H} = \{$ 2차원 평면 위 직선들 $\}$

$\Rightarrow VC(\mathcal{H}) = 3$



점 3개

임의의 점 4개는 $\mathcal{H}$로 shatter 할 수 없음.

- $D = VC(\mathcal{H})$

- growth func : $\Pi_{\mathcal{H}}(n) = \max_{x_1 \cdots x_n} \left| \{ (h(x_1) \ldots h(x_n)) : h \in \mathcal{H} \} \right|$ i.e $\mathcal{H}$가 만들 수 있는 label의 개수

- Sauer's lemma : $\Pi_{\mathcal{H}}(n) \le \sum_{i=0}^{D} \binom{n}{i} \le \left(\frac{en}{D}\right)^D$

- $Pr(\exists h \in \mathcal{H} : |\varepsilon(h) - \hat{\varepsilon}(h)| > r) \le 2 \left(\frac{en}{D}\right)^D \exp(-2nr^2)$

- with prob $\ge 1-\delta$

$\because \delta = 2\left(\frac{en}{D}\right)^D \exp(-2nr^2)$

(i) $|\varepsilon(h) - \hat{\varepsilon}(h)| \le \sqrt{\frac{D\log(en/D) + \log(2/\delta)}{2n}}$

$r = \sqrt{\frac{D\log(en/D) + \log(2/\delta)}{2n}}$

(ii) $\varepsilon(\hat{h}) \le \varepsilon(h^*) + 2\sqrt{\frac{D\log(en/D) + \log(2/\delta)}{2n}}$

$\because \varepsilon(\hat{h}) \le \hat{\varepsilon}(\hat{h}) + r$

$\le \hat{\varepsilon}(h^*) + r \le \hat{\varepsilon}(h^*) + 2r$

- for $|\varepsilon(h) - \hat{\varepsilon}(h)| \le r$

to hold, with prob $\ge 1-\delta$

$n \ge \frac{1}{2r^2}\left(D\log\frac{en}{D} + \log\frac{2}{\delta}\right)$

Note.

D linear to # of params

$\Rightarrow$ n linear to # of params

---

1. Regularization

$J_\lambda(\theta) = J(\theta) + \lambda R(\theta)$ where

(i) $R(\theta)$: regularizer (non-negative)

(ii) $J_\lambda(\theta)$: regularized loss

(iii) λ : regularization parameter

- L2 $R(\theta) = \frac{1}{2}\|\theta\|_2^2$

  : $\theta \leftarrow \underbrace{(1-\eta\lambda)}_{\text{weight decay}}\theta - \eta \nabla J(\theta)$

- L0 $R(\theta) = \|\theta\|_0$ i.e # of parameter

  ⇒ 미분 불가능

  ⇒ L1 $R(\theta) = \|\theta\|_1$ (LASSO)

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2. Implicit Regularization

Optimization

⇒ classical : 1 global minimum

⇒ deep learning : 多 " "

tip (i) large init learning rate

(ii) smaller init

(iii) smaller batch size

(iv) momentum.

3. Cross Validation

- Naive
  - models $M = \{M_1, \ldots, M_d\}$

    $\Big\downarrow$ training set S
  - hypothesis $\mathcal{H} = \{h_1, \ldots, h_d\} \Rightarrow$ minimize $\hat{\varepsilon}_S(h)$

- Hold-Out CV
  - models $M = \{M_1, \ldots, M_d\}$

    $\Big\downarrow$ training set S_{train}
  - hypothesis $\mathcal{H} = \{h_1, \ldots, h_d\} \Rightarrow$ minimize $\hat{\varepsilon}_{S_{cv}}(h)$
  - $|S_{train}| = 70\%$ $|S_{cv}| = 30\%$

- K-Fold CV
  - K개 분할한뒤 $\Rightarrow$ K번 학습 (1조각만 결과용으로)

    $\Rightarrow$ 가장 낮은 평균 오차 모델 M 선택

    $\Rightarrow$ 전체 데이터 S로 재학습.
  - leave-one-out CV : $K = \#$ of data.

Frequentist View

, $\theta_{MLE} = \arg\max_{\theta} \prod_{i=1}^{n} p(y^{(i)} | x^{(i)}; \theta)$

Bayesian View

: $p(\theta | S) = \dfrac{p(S|\theta) \cdot p(\theta)}{p(S)}$

MAP

: $\theta_{MAP} = \arg\max_{\theta} \prod_{i=1}^{n} p(y^{(i)} | x^{(i)}, \theta) \cdot p(\theta)$

Note. $\theta \sim N(0, \tau^2 I)$

with θ_{MAP} smaller norm than θ_{MLE}

$\Rightarrow$ less overfitting

- let $p(y^{(i)} | x^{(i)}, \theta) = \sigma^{y^{(i)}} (1-\sigma)^{1-y^{(i)}}$

  with $\sigma = \text{sigmoid}(-\theta^T x^{(i)})$

- $p(y | x, S) = \int_{\theta} p(y | x, \theta)\, p(\theta | S)\, d\theta$

- $E[y | x, S] = \int_{y} y\, p(y | x, S)\, dy$